

BIGKAS — MASTER PROPOSAL

Create & Conquer 2026 · Theme 2 (IBM-sponsored) · The missing assessment-data layer for foundational literacy

1. The Problem

Theme 2 asks us to fix the broken path from a known problem to a solution that can scale. Philippine learning poverty is severe and precisely measured. In 2022, the World Bank recorded a learning poverty rate of 91% — meaning nine in ten ten-year-olds cannot read and understand an age-appropriate text, a rate 56.4% worse than the East Asia and Pacific average. A 2024 government survey corroborates this: basic literacy stands at 93.1%, but functional literacy — reading with comprehension — sits at only 70.8%, meaning roughly two in every nine literate Filipinos cannot adequately understand what they read. (<https://southeastasiaglobe.com/philippines-falls-behind-in-childrens-learning/> , <https://www.gmanetwork.com/news/topstories/nation/954393/psa-only-70-8-of-filipinos-aged-10-64-functionally-literate/story/> , <https://edcom2.gov.ph/student-proficiency-rates-plunge-from-30-in-grade-3-to-0-47-in-grade-12/> , <https://newsinfo.inquirer.net/2235899/solon-depeds-focus-should-on-students-reading-not-other-policies>)

The path breaks in two places:

In the classroom: teachers cannot efficiently identify which child is struggling and why. The official Phil-IRI reading test is conducted orally, one child at a time, by a teaching force of 875,514 — already short by approximately 56,050 — across 45,199 public schools. National reading data consequently never translates into targeted help for individual children. ([Manila Bulletin](#))

In the system: there is no fast, detailed, or comparable reading data at scale. The real figure of 91% is nearly three times the government's own estimate of approximately 37%. Across five surveyed regional countries, officials consistently overestimated children's literacy levels by large margins — decision-makers are operating on a picture roughly one-third the size of the actual problem. (<https://businessmirror.com.ph/2023/09/22/91-learning-poverty-thrice-govt-estimates/>)

The World Bank's recommended starting point is direct: "assess learning." That is precisely the step that cannot currently be done at scale. Bigkas automates it. ([World Bank](#) [1])

2. Executive Summary

Bigkas is a bilingual reading assessment tool powered by IBM Granite Speech. A child reads a short, grade-level passage aloud for approximately 60–90 seconds. Granite Speech transcribes the reading and captures word-level timing, enabling Bigkas to measure reading fluency. Immediately after, a brief comprehension check determines whether the child understood the passage. Both assessments run

independently per language, so a teacher can identify, for example, that a child reads adequately in Filipino but struggles in English.

From these results, Bigkas generates a per-child report identifying specific skill gaps, automatically groups the class into reading tiers with corresponding drills, sends a short parent message in the family's language, and aggregates results from the classroom up to the school, division, and national level.

Bigkas addresses the critical missing step in Theme 2's pathway: the gap between identifying a child's reading difficulty and delivering a targeted response at scale. It gives school systems their first structured, school-level map of where reading is failing. The application runs offline on low-cost Android devices and improves over time through a data flywheel that progressively builds a fine-tuned Filipino reading model.

3. Why Now (Market & Policy Timing)

Three conditions converge in 2026. First, AI-driven assessment has matured — automated assessment and the automation of routine teaching tasks are identified as primary EdTech growth drivers, with K-12 as the largest market segment. Second, DepEd is actively digitizing operations, distributing laptops and Smart TVs and streamlining enrollment processes. Third, foundational literacy has become a top national policy priority following EDCOM 2. The tool, institutional readiness, and political will are simultaneously present. ([Grand View Research \[8\]](#); [Manila Bulletin \[4\]](#);

<https://edcom2.gov.ph/student-proficiency-rates-plunge-from-30-in-grade-3-to-0-47-in-grade-12/>)

4. The Solution / Innovation (*detailed*)

Bigkas measures the two core components of reading — fluency and comprehension — in both Filipino and English, separately, and converts those results into a prioritized lesson plan for the teacher and a live, school-by-school national map of where reading is failing.

4.1 The 7-Step Assessment Loop

Think of it as one smooth loop the teacher runs in class:

Step 1 — Oral reading capture. The student reads a grade-level passage (approximately 60–120 words) displayed on the teacher's phone. The app records the reading offline and syncs later. Each assessment is single-language — Filipino and English passages are never mixed — so a child pausing to translate a word is not misclassified as a reading error. Passages are long enough that the child is still reading at the one-minute mark, ensuring a valid measure of reading rate.

Step 2 — Fluency scoring via Granite Speech. Granite Speech transcribes the recording and captures word-level timestamps. Bigkas aligns the transcription against the target passage and scores four fluency sub-skills: decoding and phonics, syllable blending, sight word recognition, and pacing and accuracy. Each error is tagged to a sub-skill, so the pattern of errors identifies which skill is weak — not merely how many errors occurred.

Step 3 — Comprehension check. Immediately after the reading, Bigkas presents 4–5 short questions, each mapped to a distinct comprehension sub-skill: literal recall, vocabulary in context, inference, and sequencing or main idea. The child taps or speaks the answer. Each incorrect response points to a specific comprehension gap.

Step 4 — Per-child report. The teacher receives a structured report showing fluency performance, specific words the child struggled with, comprehension results by sub-skill, and an overall reading level — for each language assessed.

Step 5 — Automatic class grouping with skill rationale. Students are sorted into three tiers. The color indicates urgency; the attached skill profile identifies what broke and what to address first.

- **Green — On Track:** strong fluency and comprehension. Assigned enrichment activities.
- **Yellow — Needs Practice:** one strand is weak. Splits into Yellow-Fluency (comprehension intact but reading is slow or inaccurate → fluency drills) and Yellow-Comprehension (fluent reading but poor understanding → comprehension drills). Same color; opposite interventions.
- **Red — Needs Support:** both strands are weak, typically with a foundational decoding gap at the root. Intervention begins at phonics and syllable blending, as comprehension cannot improve until decoding is stable.

Because Filipino and English are scored independently, a child may be Green in Filipino and Red in English, and the teacher sees both profiles.

Step 6 — Parent SMS. A short message in the family's language describes the child's reading status and provides one actionable home-support suggestion.

Step 7 — System-wide rollout. Results aggregate from class to school to division to national level, replacing a single national statistic with a school-by-school, skill-by-skill, language-by-language map of where reading is failing.

4.1.1 Grouping Logic — The Reading-Ladder Rule

Reading is a developmental ladder; the lowest broken rung must be fixed first. When a foundational rung fails, higher rungs typically appear broken as downstream symptoms rather than independent problems.

None

FLUENCY

COMPREHENSION

- | | | |
|-----------------------|---|---------------------------|
| 1. Decoding / phonics | → | 4. Literal recall |
| 2. Syllable blending | → | 5. Vocabulary in context |
| 3. Sight words | → | 6. Inference |
| (Pacing / accuracy = | | 7. Sequencing / main idea |

aggregate signal of 1–3)

The rule runs independently for Filipino and English:

1. Identify the lowest broken rung in each strand.
2. Rate severity by rung position: a break at decoding or blending is severe; at sight words, inference, or sequencing is moderate; performance below the pacing or accuracy benchmark without a broken rung is mild; no break is solid.
3. Assign the tier: both strands solid → Green; one strand has a gap → Yellow (labeled Fluency or Comprehension); both strands have gaps, or either shows a severe foundational gap → Red.
4. Set the primary drill to the lowest broken rung. Remaining gaps are flagged as "likely downstream — re-check after intervention," and a re-test is scheduled.

All thresholds follow published DepEd and Phil-IRI grade-level benchmarks.

4.2 A concrete example (so anyone can picture it)

The following is a constructed example for demonstration purposes, not real assessment data.

In Aling Rosa's Grade 2 class, Juan reads: "Si Ana ay may maliit na pusa. Puti ang pusa at mahilig maglaro." He skips "maliit," substitutes "puli" for "puti," and pauses before "mahilig." His results:

- **Filipino** — ● **Yellow (Comprehension)**. Fluency is solid (WCPM 62, accuracy 94%). The lowest broken comprehension rung is inference — literal recall and vocabulary are intact. → **Primary drill: inference and retelling.**
- **English** — ● **Red (Needs Support)**. The lowest broken rung is decoding (severe), accounting for weak blending, slow pacing, and poor comprehension above it. → **Primary drill: English phonics and decoding. Remaining gaps flagged as likely downstream — re-check in two weeks.**

One child. Two languages. Two prioritized action plans. The teacher receives two root causes in priority order, not seven separate problems.

4.3 Innovation, Part 1 — Closing the Full Loop

Most reading tools produce a score and stop — no next step for the teacher, no system-level data. Manual Phil-IRI provides a diagnosis but consumes substantial teacher time and generates no aggregated output. Foreign reading applications do not support Filipino. Bigkas owns the complete chain in a single loop — assess, diagnose, group, intervene, notify parents, and roll up — in two languages. The distinction is between "here is a score" and "here is tomorrow's lesson, differentiated by need, alongside the first real picture of where the country's reading is breaking."

4.4 Innovation, Part 2 — Keyword Biasing

The problem. Recognizing a child reading in Filipino is difficult. Open-domain ASR on clean adult Tagalog audio achieves approximately 78–85% accuracy, and performance degrades further with Taglish code-switching, regional accents, and children's voices. (ConvertAudioToText [9])

The insight. In a reading assessment, the expected output is already known — it is the passage on screen. The model does not need to transcribe arbitrary speech; it needs to verify how closely the child's reading matches a known text. This is the difference between a dictation test — transcribe an unknown speaker with no reference — and a spelling-bee judge holding the word list. Bigkas is the judge with the list.

How Granite makes it work. IBM Granite Speech supports keyword biasing: passage words are supplied as input, and the model is nudged toward recognizing them correctly. Word-level timestamps enable precise measurement of reading speed and pauses. This reframes a hard open-transcription problem as a tractable alignment problem. Granite Speech is open-source (Apache 2.0) and compact, making later fine-tuning on Filipino children's speech both feasible and the foundation of a lasting competitive moat. (IBM [10]; DEV Community [11])

4.5 Why AI Is Necessary

Manual Phil-IRI requires a teacher to assess each child individually — approximately 10–15 minutes per child, scored by ear. For a class of 40 or more students, this spans many hours across multiple days, during which the teacher cannot simultaneously instruct the class. A conventional application cannot substitute for this process because the core task — evaluating a child's spoken reading — requires speech AI. With Bigkas, the child reads to the device, the AI scores the reading, and the teacher's active involvement is minimal. Each child requires approximately 2–3 minutes on the device and can rotate through a station while the rest of the class continues working. The result is uniform, objective scoring for every child in both languages, with hours previously spent on manual testing returned to instruction.

4.6 Alignment with Theme Requirements

Theme Word	Requirement	How Bigkas Delivers
Context-aware	Fits local language, setting, and technology	Filipino-first assessment with separate English scoring; offline operation on low-cost Android devices; workflow aligned to Phil-IRI

Immediately implementable	Functional in real conditions today	Runs on the teacher's existing phone; no new hardware or infrastructure required
Scalable	Grows beyond the initial site	Software scales to millions of users; expansion via additional language packs; data flywheel compounds over time
Effective	Produces measurable results	2–3 minutes per child with near-zero teacher scoring time; uniform scoring; specific sub-skill diagnosis per language

4.7 Summary Explanations at Three Levels

10 seconds: A child reads aloud for about a minute, answers a few quick questions, and the AI immediately tells the teacher which reading skill that child is missing — in Filipino and English.

30 seconds: Nine in ten Filipino ten-year-olds cannot read and understand a simple text, yet the government estimates the figure at three in ten. Bigkas enables any teacher to assess a child's reading fluency and comprehension in both languages using IBM Granite Speech, identifies the specific weak skill, sorts each child into a reading group with a targeted drill, and aggregates every result into the first school-by-school map of where Philippine reading is failing.

2 minutes: The walkthrough in Section 4.1 + the ladder logic in Section 4.1.1 + the keyword-biasing explanation in Section 4.4.

4.8 Languages Assessed

Philippine classrooms are formally bilingual: early grades use Filipino or the mother tongue as the primary medium, with English introduced progressively. Bigkas runs two independent assessments — one in Filipino, one in English — each benchmarked separately. Languages are never mixed in a single passage, as Taglish input would make it impossible to distinguish a genuine reading error from a translation attempt. Note: Granite Speech officially supports English, French, German, Spanish, Portuguese, and Japanese — not Filipino. The Filipino assessment currently relies on keyword biasing and is designed for later fine-tuning on Filipino children's speech data.

4.9 Honest Framing

Bigkas is a fast bilingual screener, not a clinical diagnostic instrument. All performance thresholds follow published DepEd and Phil-IRI benchmarks. Reliability is established through repeated assessment across a term and trend observation, not from any single 2–3 minute session.

5. Unique Value Proposition and Competitive Advantage

UVP: Fast, bilingual reading assessment — fluency and comprehension, in Filipino and English — that identifies the specific weak skill, tells the teacher what to address tomorrow, and gives the system its first structured map of where reading is failing.

The moat: Bigkas combines bilingual assessment, spoken input, comprehension measurement, sub-skill-level diagnosis, an integrated action layer, child-safe design, and full-loop coverage in the Philippine context. Competing tools address one or two of these elements; none close the complete loop. The durable competitive advantage is the data flywheel: increased usage generates Filipino children's speech data, which enables model fine-tuning, which improves product accuracy, which drives further adoption.

6. Feature Breakdown

Core (PoC; full fluency loop built if advancing):

Offline reading capture · Granite Speech fluency scoring (ASR with keyword biasing, word alignment, WCPM, miscues tagged to sub-skills) · comprehension check (4–5 skill-tagged questions, tap or spoken) · bilingual assessment in Filipino and English, scored independently · ladder-based grouping into Green, Yellow-Fluency, Yellow-Comprehension, and Red · per-child report · automatic class grouping with drills · vernacular parent SMS · offline capture and background sync.

Phase 2 (post-hackathon):

Fine-tuned multi-dialect Granite model · longitudinal progress tracking · class analytics dashboard · adaptive AI-generated comprehension questions (Granite LLM) · Phil-IRI-style report export.

Long-term (full rollout):

Multi-tenant division, region, and national dashboards · role-based access, SSO, and audit logs · watsonx.governance for bias monitoring · early-warning flags for at-risk students · passage-bank authoring · DepEd LIS and LMS integration · on-premises and sovereign-cloud deployment via Red Hat OpenShift.

Build note: In the MVP, comprehension questions are authored in advance per passage for reliability. Phase 2 upgrades to AI-generated questions. This keeps comprehension fully in scope without introducing model risk.

7. Module Hierarchy

User and Identity · Roster and Organization · Assessment · AI and Speech (Granite Speech: ASR with keyword biasing, alignment and timing, miscue detection, reading-level classification, ladder and grouping engine, comprehension questions, Granite Guardian safety layer) · Intervention and Content · Notifications and Messaging · Analytics and Reporting · Dashboards · Governance and Compliance (consent management, data retention, audit logging, bias monitoring) · Sync and Offline · Integration and API · Platform and Infrastructure.

The PoC demonstrates the complete system as connected screens. If advancing, the full build covers Assessment, AI and Speech, Intervention, and the App; remaining modules are mocked.

8. System Architecture

High level: Offline-first mobile app ↔ API gateway ↔ backend services ↔ AI layer (Granite Speech and Guardian on watsonx.ai or OpenShift) ↔ data stores. watsonx is built on Red Hat OpenShift and supports on-premises or any-cloud deployment, directly addressing the government requirement to keep children's data local. (BizTech Magazine [12])

Hackathon scope: Elimination requires only a PoC; the architecture is presented as a diagram in the brief. If advancing, the system is built as a modular monolith with a clean boundary around the AI service.

Data flow: App → gateway → assessment service (audio to file storage, metadata to database) → Granite scoring → results to database → analytics → dashboards.

Key tables: organizations (type, parent), users, students, classes, passages (reference_text, grade, language), assessments (audio_uri, language, status), assessment_results (language, WCPM, accuracy, fluency_subskills JSONB, comprehension_subskills JSONB, fluency_gap_level, comprehension_gap_level, primary_drill_skill, tier, model_version), interventions, consents, audit_log, model_runs. The language field ensures Filipino and English results are stored as separate rows per child.

API: Versioned REST with authentication and role-based access. Endpoints: create assessment, upload audio, retrieve result, retrieve class grouping, retrieve organization analytics, save consent, async scoring webhook.

AI workflow:

None

Child reads aloud → Offline capture → [queue if offline]

- Granite Guardian (input safety check)
- Granite Speech ASR ◀— keyword bias = passage words; word-level timestamps
- Align vs known passage → matched / missed / substituted words + timing
 - Fluency scoring → WCPM, accuracy, miscue sub-skills (decoding, blending, sight words, pacing)
- Comprehension check → 4–5 skill-tagged questions (literal, vocabulary, inference, sequencing)
- Identify lowest broken rung per strand; rate severity (solid / mild / moderate / severe)
- Combine strands → Green / Yellow (named) / Red
- Primary drill = lowest broken rung; flag rest as "likely downstream"; schedule re-test
- Granite Guardian (output safety check)
- Per-child report · Parent SMS · Rollup: class → school → division → national
- (Full pipeline runs independently for Filipino and English)

9. IBM Granite Speech Integration

Granite Speech is a compact, open-source (Apache 2.0) speech-to-text model available on Hugging Face and watsonx.ai. The 1B parameter version ranked first on the OpenASR leaderboard at approximately 5.52% word error rate while being 50% smaller than prior-generation models. It supports keyword biasing to prioritize recognition of expected words and provides word-level timestamps. Granite Guardian is deployed at both pipeline endpoints as a risk-detection layer grounded in the IBM AI Risk Atlas, serving as the primary child-safety control.

Honest disclosure: Granite's officially supported languages are English, French, German, Spanish, Portuguese, and Japanese — not Filipino. The English assessment is fully supported natively and is safe for a live demonstration. The Filipino assessment currently relies on keyword biasing and is slated for fine-tuning on Filipino children's speech as usage scales — constituting the primary data moat. Audio clips of 60–90 seconds process efficiently on the 1B and 2B models; background and batch scoring ensure latency does not affect classroom usability or drive up compute costs. (IBM [10]; DEV Community [11])

10. Dashboard Decision — Option C (Final)

Full working app + Analytics Dashboard (real UI on seeded data) + Admin Portal (shell with key screens mocked). The analytics dashboard is a core product deliverable — the system-level value proposition is incoherent without a visible map. The admin portal is necessary for production but does not differentiate the product for judges and is therefore mocked at the shell level.

11. Stakeholder Analysis

Stakeholder	What They Receive
Teachers (primary)	Instant, fair, bilingual diagnosis with a ready lesson plan; manual testing time recovered
Students and parents (secondary)	Targeted academic support; simple home guidance in the family's language
School heads and supervisors	Class- and school-level reading view by language; ability to direct support where needed
DepEd and literacy NGOs	System-level reading data not currently available; SDG-4 reporting support; donor-ready results
Private school networks and publishers	Assessment-as-a-service; diagnostic capability integrable into existing programs
DepEd national and division offices, LGUs	Anchor buyer; data infrastructure for policy decisions, resource targeting, and accountability

12. Market Model

The market. The Philippine EdTech market was approximately USD 5.0 billion in 2024 and is projected to reach approximately USD 13.65 billion by 2033 at a CAGR of roughly 11.8%, within a total Philippine education market of approximately USD 31.2 billion serving over 28 million basic education students. Global EdTech reached approximately USD 187 billion in 2025, with K-12 as the largest segment and AI-driven assessment as a primary growth driver. (IMARC [6]; Nexdigm [7]; Grand View Research [8])

TAM / SAM / SOM (indicative; pricing at ₱100/student/year is an estimate subject to validation):

Tier	Scope	Students	Approximate Annual Value
TAM	All public basic education learners in reading grades	~27.6M (~15.4M elementary), SY 2025–26	~₱2.7B (~USD 48M)
SAM	Public Grades 1–3 in areas with adequate Android access and connectivity	~4–6M	~₱400–600M (~USD 7–11M)
SOM (3-year)	Pilot divisions, NGO clusters, and early private network adopters	~0.3–1M	~₱30–100M (~USD 0.5–1.8M)

User personas. Teacher Maria — Grade 2, 45 pupils, limited time, skeptical of new tools — wins when Bigkas delivers tomorrow's differentiated lesson plan with no additional scoring work. Superintendent Reyes needs reading data to justify budget allocations and target support. The NGO program lead needs measurable reading-gain results for donor reporting. The private school director will pay a premium for integrated assessment-as-a-service.

Competitive landscape.

Type	Examples	Why Bigkas Wins
PH EdTech and LMS	Quipper, Edukasyon.ph, 51Talk, Edusuite, Rex Education	Content and LMS focus; none support spoken bilingual fluency assessment with comprehension and an action layer

Foreign reading apps	Google Read Along, Lexia, Amplify	No Filipino or Taglish support; assume device access and bandwidth that rural Philippine schools lack
Status quo	Manual Phil-IRI	One child at a time, scored by ear, with no system-level data output

Adoption barriers. Over half of Filipino teachers prefer face-to-face instruction, and digital literacy gaps slow EdTech uptake. Device and connectivity shortfalls in underserved schools are additional constraints. Mitigations: offline-first design, Phil-IRI-aligned output, brief teacher onboarding, and NGO or division champions as early adopters. (Ken Research [5])

Market entry and growth. Phase 1: one NGO or division design partner, one structured pilot, one hard reading-gain metric. Phase 2: publish results and expand to division-wide rollouts and private school pilots. Phase 3: regional scale, DepEd national agreement, additional language packs (Bisaya, Ilocano), and adjacent international markets. Initial deployment targets Tagalog-speaking areas for the strongest baseline ASR accuracy. The data flywheel accelerates at each stage: more usage → more Filipino children's speech data → model fine-tuning → improved performance → further adoption.

13. Business Model

Revenue streams.

1. Government licensing — per-student or per-school annual fee; national framework agreements
2. NGO and donor-funded rollouts — project-based fees tied to program outcomes
3. Freemium — free for a single classroom; paid above that threshold
4. Assessment-as-a-service — premium tier for private school networks and publishers
5. Long-term: anonymized, consented benchmark data licensed for policy research

Indicative pricing (*estimates subject to market validation*):

Segment	Model	Approximate Price	Rationale
Public schools	Per student or per school per year	₱60–120/student or ₱15k–40k/school	Typically grant-funded; volume-driven

NGO and donor programs	Project fee plus per-student rate	Set per program	Funded against measurable outcomes
Private networks	Per student per year (premium)	₱200–400/student	Greater willingness and ability to pay
Individual teachers	Freemium → paid	Free for one class	Bottom-up adoption funnel

Cost structure. Primary costs are compute for scoring (kept low through small Granite models, batching, and edge processing), language and content development, engineering and support, hosting, and sales. Marginal cost per additional student is near zero, so unit economics improve significantly at scale.

Unit economics. Government contracts carry high acquisition costs but high lifetime value across multi-year, multi-school agreements. The ratio improves once a strong reference site exists. NGO and private channels have lower acquisition costs and faster revenue cycles, bridging the extended government sales timeline.

Funding path. Hackathon prize → grant and CSR funding for pilots → private-network revenue to bridge → government agreements for national scale. For reference, EDGE Tutor raised USD 1M pre-Series A in February 2025, indicating early-stage EdTech funding in this market is achievable. (IMARC [6])

14. Impact Model

At the teacher level, replacing a 10–15 minute manual Phil-IRI session per child with a 2–3 minute device-administered assessment returns hours per assessment cycle directly to instruction. At the system level, replacing a single national statistic with a school-by-school, skill-by-skill, language-by-language map enables targeted resource allocation — the practical difference between knowing "91% of children cannot read with comprehension" and knowing "these 40 schools, this skill, this language, this quarter." Primary tracking metrics: share of students advancing one reading group per term, and time elapsed between assessment and intervention. Long-term objective: identifying and addressing reading gaps before age ten, while remediation remains most effective.

15. Analytics and KPIs

North-star metric: Students advancing one reading group per quarter.

Supporting KPIs: Assessment coverage rate · Time-to-intervention · Reading gain by group · Equity-gap closure by region, gender, and language · Teacher adoption rate.

Dashboard layers: Teacher · School · Division · National.

Report types: Phil-IRI-style exports · Donor and program reports · Equity and disaggregated reports.

Assessment data without an action layer produces no change. Bigkas closes the loop by converting results into decisions about where to direct help — and by making that chain visible at every level of the system.

21. References

1.

World Bank (2022), *Fixing the Foundation* — Philippine learning poverty at 91%, 56.4% worse than EAP; "assess learning."

Southeast

Asia

Globe

—

<https://southeastasiaglobe.com/philippines-falls-behind-in-childrens-learning/>

World

Bank

Learning

Poverty

Brief

(PDF)

—

<https://documents1.worldbank.org/curated/en/099090524085540633/pdf/P179209-b0057b3a-fd8c-476a-a1c6-271b276117ff.pdf>

Philstar

—

<https://www.philstar.com/headlines/2023/09/22/2298233/learning-poverty-philippines-linked-poor-teaching-quality-world-bank-study>

2.

BusinessMirror — 91% learning poverty is nearly three times the government's ~37% estimate.

<https://businessmirror.com.ph/2023/09/22/91-learning-poverty-thrice-govt-estimates/>

3.

PSA — 2024 FLEMMS: 93.1% basic literacy, 70.8% functional literacy.

GMA

News

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<https://www.gmanetwork.com/news/topstories/nation/954393/psa-only-70-8-of-filipinos-aged-10-64-functionally-literate/story/>

PSA

—

<https://psa.gov.ph/content/every-10-filipinos-9-have-basic-literacy-while-7-have-functional-literacy>

Manila Bulletin — <https://mb.com.ph/2025/5/5/dep-ed-clarifies-flemms-2024-results>

4.

DepEd / Manila Bulletin — 875,514 teachers; ~56,050 shortage; 45,199 public schools; 27.6M enrollees (15.42M elementary).

<https://mb.com.ph/2025/06/15/school-year-20252026-deped-projects-276-million-enrollees-in-basic-education>

5.

Ken Research — Philippine EdTech competitors; >50% of teachers prefer face-to-face; teacher digital-literacy gap.

<https://www.kenresearch.com/philippines-e-learning-and-edtech-market>

6.

IMARC Group — Philippine EdTech ~USD 5.0B (2024) → ~USD 13.65B (2033), ~11.8% CAGR; EDGE Tutor USD 1M pre-Series A (Feb 2025).

<https://www.imarcgroup.com/philippines-edtech-market>

7. Nexdigm — Philippine education market ~USD 31.2B; 28M+ basic-ed students; ~USD 70M EdTech funding (2023).

<https://www.nexdigm.com/market-research/report-store/philippines-education-market-research-report/>

8. Grand View Research — global EdTech ~USD 187B (2025); K-12 largest segment; AI-driven assessment a key driver.

<https://www.grandviewresearch.com/industry-analysis/education-technology-market>

9. ConvertAudioToText — Tagalog ASR ~78–85% on clean adult audio; worse with Taglish and accents.

<https://convertaudiototext.com/blog/tagalog-filipino-transcription>

10. IBM Granite — open, trusted speech and guardian models (Apache 2.0).

<https://www.ibm.com/granite>

IBM Granite 3.3 Speech announcement —

<https://www.ibm.com/new/announcements/ibm-granite-3-3-speech-recognition-refined-reasoning-rag-loras>

Hugging Face (granite-speech-3.3-8b) —

<https://huggingface.co/ibm-granite/granite-speech-3.3-8b>

11. DEV Community — Granite Speech #1 on OpenASR (~5.52% WER); keyword biasing usage.

<https://dev.to/aairom/granite-40-1b-speech-model-is-out-and-its-dynamite-2f60>

Hugging Face (granite-speech-4.1-2b, supported languages and word timestamps) —

<https://huggingface.co/ibm-granite/granite-speech-4.1-2b>

12. BizTech Magazine — watsonx built on Red Hat OpenShift; on-premises or any-cloud deployment.

<https://biztechmagazine.com/article/2025/12/ibms-watsonx-platform-goes-distance-ai-governance-financial-institutions>

13. IBM watsonx.governance — Leader, 2026 Gartner Magic Quadrant for AI Governance Platforms.

<https://www.ibm.com/products/watsonx-governance>

14. EDCOM 2 — student proficiency data.

<https://edcom2.gov.ph/student-proficiency-rates-plunge-from-30-in-grade-3-to-0-47-in-grade-12/>

15. Inquirer — DepEd focus on student reading.

<https://newsinfo.inquirer.net/2235899/solon-depeds-focus-should-on-students-reading-not-other-policies>